

The key advantages of using pre-trained models based libraries are:

- Inclusion of pre-trained weights with NLP architectures
- Inclusion of fine-tuning with pre-processing
- Easy-to-use scripts and APIs
- Multilingual support with international and regional languages
- Compatibility with graphics processing unit (GPU)
- Pre-programmed algorithms from leading companies

Installation and working with pre-trained NLP based models

HuggingFace (<https://huggingface.co/>) is one of the key platforms that provide pre-trained models for natural language processing (NLP). It is cloud based and can be integrated with Google Colab for running scripts.

To install the pre-trained NLP based models in Google Colab, execute the following:

```
! pip install pytorch-transformers
! pip install transformers
! pip install sentencepiece
```

Prediction of the next sequence in Google Search

When we write some text in Google Search, the next sequence is suggested by the back-end library of Google. For example, if we want to predict the next word after the sentence 'What is the name of the Indian?', the following transformer can be used:

```
import torch
from pytorch_transformers import GPT2Tokenizer,
GPT2LMHeadModel
mytokenizer = GPT2Tokenizer.from_pretrained('gpt2')
# Encode a text inputs
text = "what is name of the Indian "
indexed_tokens = mytokenizer.encode(text)
```

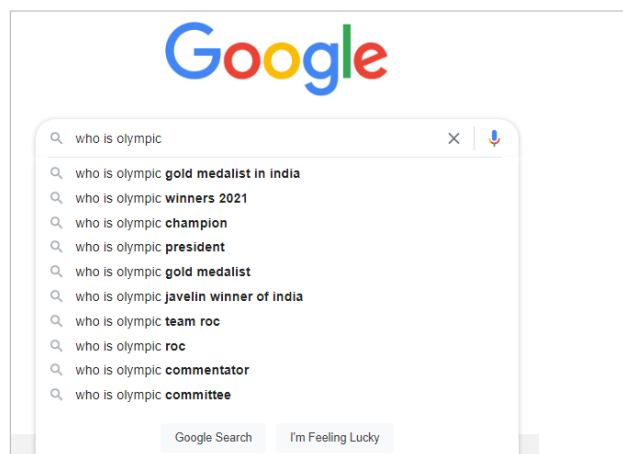


Figure 3: Prediction of the next sequence in Google Search

```
tokens_tensor = torch.tensor([indexed_tokens])
model = GPT2LMHeadModel.from_pretrained('gpt2')
model.eval()
tokens_tensor = tokens_tensor.to('cuda')
model.to('cuda')
with torch.no_grad():
    outs = model(tokens_tensor)
    preds = outs[0]
pred_index = torch.argmax(preds[0, -1, :]).item()
pred_text = tokenizer.decode(indexed_tokens + [pred_index])
print(pred_text)
```

The output from the execution of this code will be predicted depending on the following:

- flag
- parliament
- others, depending upon the search

HuggingFace is providing pre-trained models for a very wide range of applications and being used by numerous corporate giants.

More than 5,000 organizations are using Hugging Face	
Allen Institute for AI Non-Profit · 93 models	Facebook AI Company · 298 models
Graphcore Company · 25 models	Google AI Company · 492 models
Amazon Web Services Company · 1 model	SpeechBrain Non-Profit · 36 models
Microsoft Company · 149 models	Grammarly Company

Figure 4: Key organisations using *huggingface.co*

Prediction of a word when filling in the blanks

The classical case of filling in the blanks with real-time search can be solved using a pre-trained model of NLP.

Here is an example to predict the word that can be used in place of [MASK].

```
from transformers import pipeline
myprediction = pipeline('fill-mask', model='bert-base-uncased')
myprediction ("This is a [MASK].")
```

The output is:

```
[{'score': 0.03235777094960213,
 'sequence': 'this is a dream.',
 'token': 3959,
 'token_str': 'dream'},
 {'score': 0.030467838048934937,
 'sequence': 'this is a mistake.',
 'token': 6707,
 'token_str': 'mistake'},
 {'score': 0.028352534398436546,
```